

Gas Pressure, the Particle Model and Diffusion — Paper C

HARD • 50 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- A calculator is allowed, but every number on this paper is designed to work without one.
- Every numerical answer must carry a **unit**. An answer with no unit does not score the final mark.
- Explanations must be written **in terms of particles**. “The particles get bigger”, “the particles expand” and “the particles push harder because they are squashed” score nothing.
- The mark for each question is shown in square brackets on the right.
- No formula is given on this paper, and no value for atmospheric pressure.
- Several questions describe apparatus you may not have seen before — apply the ideas you already have.
- Where an explanation is worth three marks or more, the mark scheme is looking for **collisions per unit area per second**, and, wherever the temperature changes, for the force of each collision as well.

SECTION 1 – TWO CHANGES, ONE AFTER THE OTHER

1. A sealed gas syringe holds 60 cm^3 of air at a pressure of 100 kPa and a temperature of 300 K.
 - (a) The plunger is pushed in until the volume is 20 cm^3 , at constant temperature. [3]
Calculate the new pressure and explain the result in terms of particles.
 - (b) The syringe is then locked at that volume and heated to 600 K. Calculate the pressure now. [1]
 - (c) State one assumption about the syringe that must be true for your answer to (b). [1]

- 2.** A sealed glass flask of gas is at a pressure of 100 kPa and a temperature of 300 K. It is cooled to 150 K. The flask does not change size.

(a) State and explain what happens to the pressure, in terms of particles. [2]

(b) Calculate the new pressure. [1]

(c) A student says the pressure falls “because cooling makes the particles smaller, so there is more space”. Explain what is wrong with this and give the correct reason. [2]

SECTION 2 – THE DIFFUSION TUBE

- 3.** A 60 cm glass tube has a cotton wool plug soaked in ammonia solution at the left-hand end and one soaked in concentrated hydrochloric acid at the right-hand end. Both are put in at the same moment. After several minutes a white ring of solid forms 36 cm from the ammonia end.



(a) Explain why a ring forms at all, and name the solid that forms it. [2]

(b) Calculate how far the hydrogen chloride travelled, and the ratio of the two distances. [2]

- (c)** Explain the ratio you calculated, in terms of the particles of the two gases. [2]

- 4.** The same experiment is repeated with the whole tube warmed from 20 °C to 60 °C.

- (a)** State and explain the effect on the time the ring takes to appear. [2]

- (b)** State and explain the effect on the position of the ring. [2]

SECTION 3 – THE LONG EXPLANATION

- 5.** A sealed metal can of air is left near a fire. Its volume cannot change. After a while the can bursts. [6]

Explain fully, in terms of particles, why the pressure inside rises as the can gets hotter, and why it eventually bursts. Write in continuous prose.

- 6.** A gas at a pressure of 250 000 Pa acts on one face of a piston of area 0.008 m². The atmosphere, at 100 000 Pa, acts on the other face.

(a) Calculate the force the trapped gas exerts on the piston. [1]

(b) Calculate the force the atmosphere exerts on the piston. [1]

(c) Calculate the resultant force on the piston and state its direction. [1]

(d) The piston is held still. State the size and direction of the force holding it. [1]

End of paper. Check every answer has a unit or a form the question actually asked for.